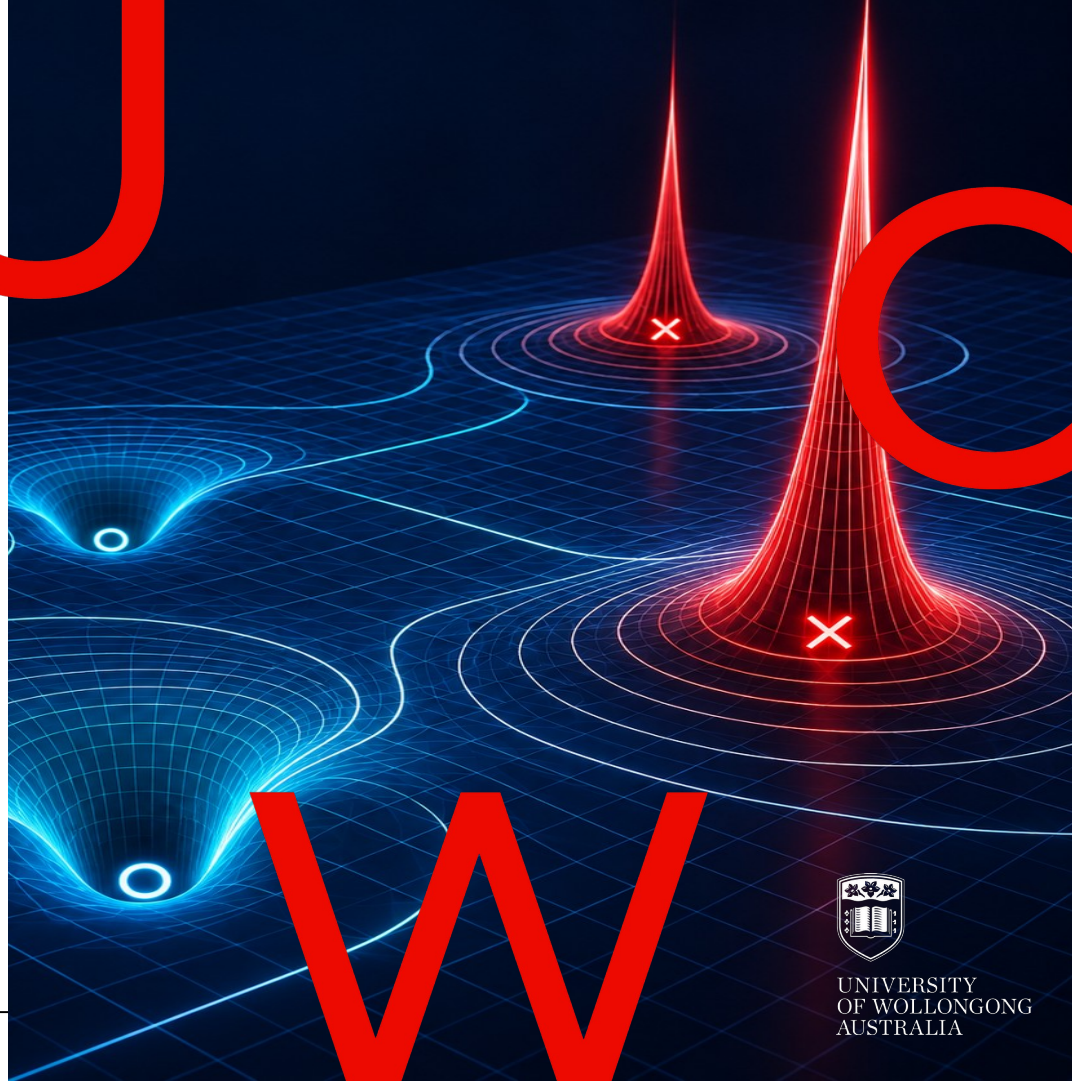
# MECH321
# Dynamics of Engineering Systems

**Lecture 4**
**Transient Response**

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PART 1

# First- and second-order time responses

01 First-order step and ramp

02 Poles and response measures

03 Full second-order step

# First-order step response

*unit step input*

$$u(t) = 1, t \geq 0 \quad U(s) = \frac{1}{s}$$

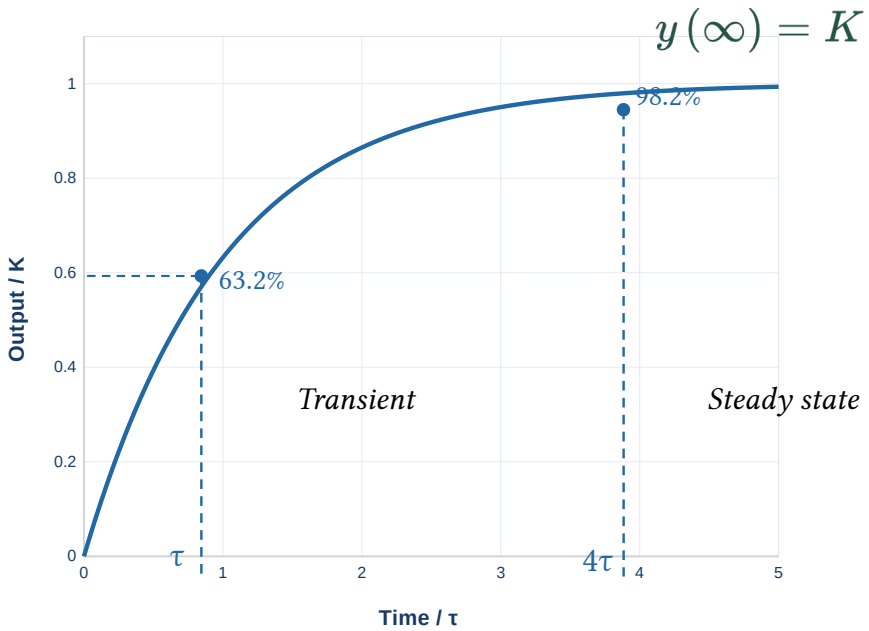
*First order system*

$$G(s) = \frac{K}{\tau s + 1}$$

$$Y(s) = G(s)U(s) = \frac{K}{s(\tau s + 1)}$$

$$y(t) = K\left(1 - e^{-t/\tau}\right)$$

$$y(\tau) = 0.632K \qquad y(4\tau) = 0.982K$$




**DC gain $K$** : final output per unit constant input.

**Time constant $\tau$:** time to complete 63.2% of the initial-to-final step change.

# Comparing two first-order systems

*Same DC gain and unit step, different time constants*

**BLUE System A**

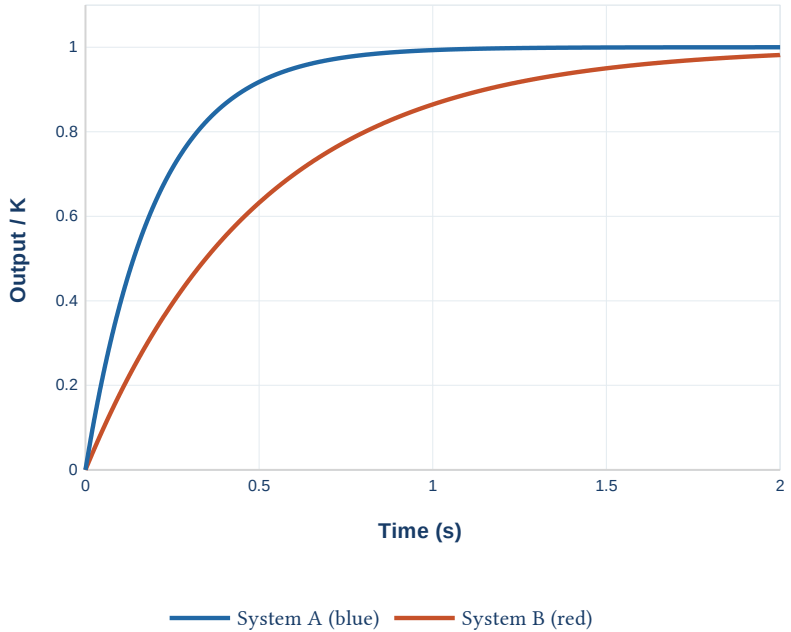
$$G_A(s) = \frac{K}{0.20s + 1}$$

$\tau_A = 0.20\,\text{s}, \quad 4\tau_A = 0.80\,\text{s}$

**RED System B**

$$G_B(s) = \frac{K}{0.50s + 1}$$

$\tau_B = 0.50\,\text{s}, \quad 4\tau_B = 2.00\,\text{s}$




**Same final value; smaller time constant gives a faster response.**

# Live demo: time constant and step response

OPEN LIVE DEMO ↗




START $K = 1$ , $\tau = 1$ s. Choose Unit step.

1 Try $\tau = 0.5$ , 1 and 2 s. Find the pole, 63.2% time and settling time.

2 Restore $\tau = 1$ s. Set $K = 2$ . Does the pole move?

MATLAB parameter: $K$ or time constant.

Course home: Demo 1. MATLAB uses 10–90% rise time here.

**MATLAB alternative (Control System Toolbox)**

```
K=1; tau=1; s=tf('s');
G=K/(tau*s+1); t=0:0.01:10;
figure; step(G,t); grid on
figure; pzmap(G); grid on
stepinfo(G,RiseTimeLimits=[.1 .9])
```

CHECK $p = -1/\tau$. 63.2% is reached at $\tau$; 2% settling is about $3.91\tau$. $K$ changes final height, not the pole.

# Ramp response and steady-state error

Unit-gain first-order system; input $u(t) = \mathrm{Bt}, t \geq 0$

$$G(s) = \frac{1}{\tau s + 1}$$

$$e(t) = Bt - y(t) \quad = Bt - B\left(t - \tau + \tau e^{-t/\tau}\right)$$

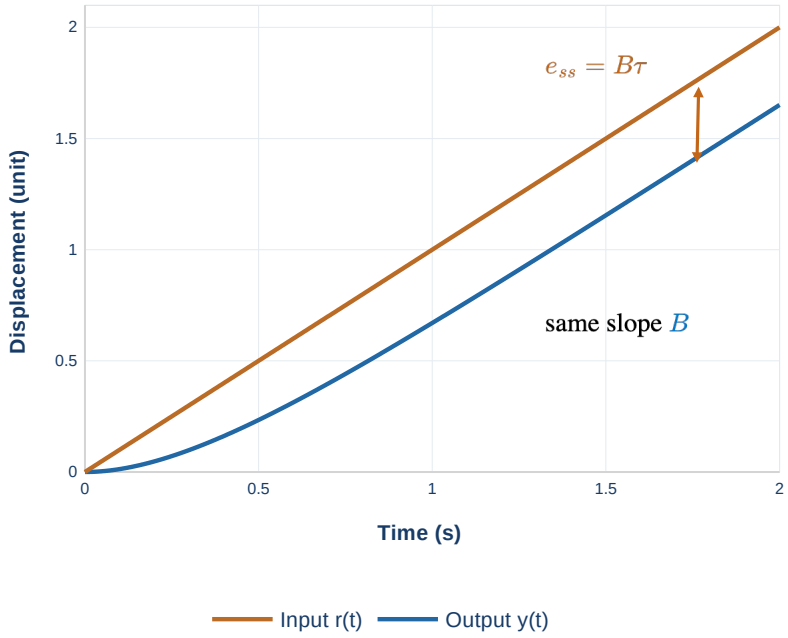
$$= B\tau\left(1 - e^{-t/\tau}\right)$$

$$e_{ss} = \lim_{t\to\infty} e(t) = B\tau$$

Ramp terms cancel; the transient vanishes.

*Using final value theorem:*

$$e_{ss} = \lim_{s\to 0} sE(s) = \lim_{s\to 0} s\frac{B}{s^2}\left(1 - \frac{1}{\tau s + 1}\right) = B\tau$$



EXAMPLE

# Example: thermometer response

*One time constant links a step test to ramp-tracking error*

A first-order thermometer reaches 98% of a step change after 8 s. The water then warms at 2 °C/s.

(a) Find $\tau$. (b) Find its steady reading error during the ramp.

$$G(s) = \frac{K}{\tau s + 1}$$

**Worked solution**

$$0.98 = 1 - e^{-8/\tau} \quad \Longrightarrow \quad \tau = \frac{-8}{\ln(0.02)} = 2.045 \text{ s}$$

$$B = 2 \text{ ° C/s} \quad \Longrightarrow \quad e_{ss} = B\tau = 4.09 \text{ ° C}$$

The $4\tau$ rule gives $\tau \approx 2$ s and error ≈ 4 °C; the values above use the exact 98% relation.

# Standard Second-Order Model



$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k} = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{b}{2\sqrt{mk}}, \quad K = \frac{1}{k}$$

**Change one physical parameter; keep the others and the force-step height fixed.**

More mass: lower natural frequency and damping ratio; slower decay; same final displacement.

More damping: less overshoot; same natural frequency and final displacement. Beyond critical damping, the slow pole moves toward zero and settling slows.

Stiffer spring: higher natural frequency, lower damping ratio and final displacement. The underdamped decay rate is unchanged: faster ringing need not mean faster settling.

# Underdamped second-order step response



$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \qquad U(s) = \frac{B}{s}, \qquad 0 < \zeta < 1$$

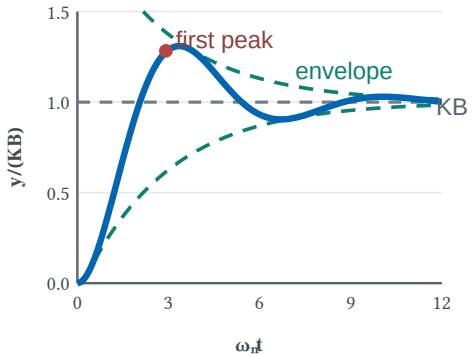
$$Y(s) = G(s)U(s) = \frac{KB\omega_n^2}{s\left(s^2 + 2\zeta\omega_n s + \omega_n^2\right)}$$

$$= \frac{KB\left(\sigma^2 + \omega_d^2\right)}{s\left((s+\sigma)^2 + \omega_d^2\right)}$$

$$Y(s) = KB\left[\frac{1}{s} - \frac{s + 2\sigma}{(s+\sigma)^2 + \omega_d^2}\right], \qquad \sigma = \zeta\omega_n, \quad \omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$= KB\left[\frac{1}{s} - \frac{s+\sigma}{(s+\sigma)^2 + \omega_d^2} - \frac{\sigma}{(s+\sigma)^2 + \omega_d^2}\right]$$

$$y(t) = KB\left[1 - e^{-\sigma t}\left(\cos\omega_d t + \frac{\sigma}{\omega_d}\sin\omega_d t\right)\right]$$

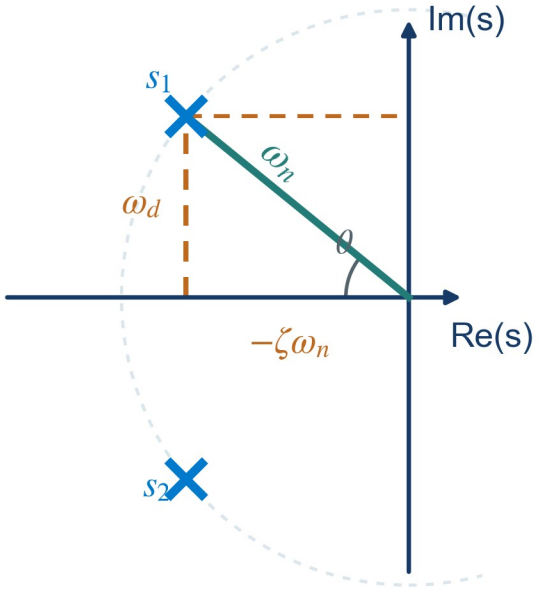



Illustrative response, damping ratio 0.35

Check: $y(0) = 0$ and $y(\infty) = KB$.
The exponential envelope falls with time constant $1/\sigma$; the sinusoid rings at $\omega_d$.

# Pole location and damping regime






Free response: same system poles.
Initial conditions set the coefficients.

$$s_{1,2} = -\zeta\omega_n \pm j\omega_d, \omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$|s| = \omega_n, -\mathrm{Re}\,(s) = \zeta\omega_n$$

The **real part sets the decay envelope**. The **imaginary part sets the ringing frequency**. Distance from the origin gives $\omega_n$.

Oscillation period: $T_d = 2\pi/\omega_d$.
Adjacent zero crossings are $T_d/2$ apart.
L3 slide 21: $\omega_d = 5$ rad/s → $T_d \approx 1.26$ s.

$0 < \zeta < 1$: complex pair and overshoot. $\zeta = 0$: sustained oscillation. $\zeta \geq 1$: real poles and no oscillatory overshoot.

# Reading a second-order step response



*Define every transient measure against the final value*

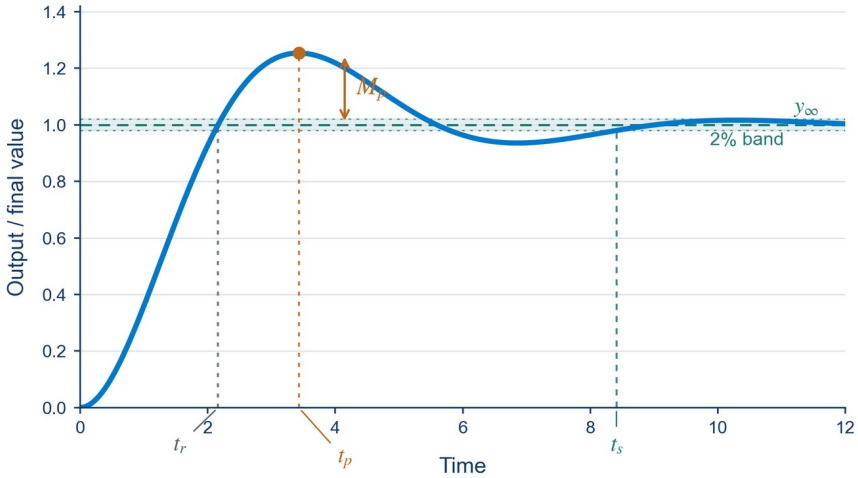
**Final value** $y_\infty = KB$

Rise time $t_r$: first arrival at $y_\infty$ (0–100% definition).

Peak time $t_p$: time of the first maximum.

Overshoot: (peak − final) / final; ×100 for percent.

Settling time $t_s$: after this time the response stays within the stated band, usually ±2% of $y_\infty$.



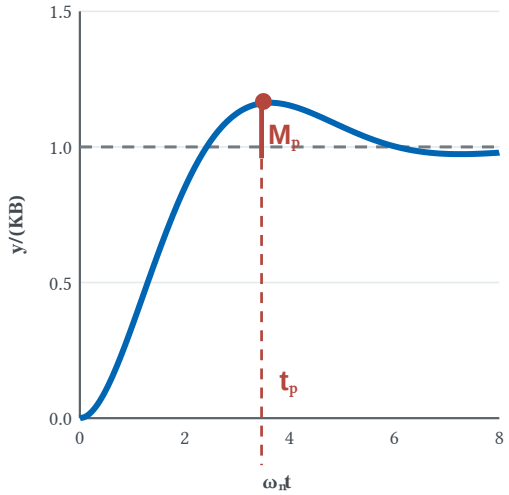
# Peak time and overshoot: derivation

CORE

$$\dot{y}(t) = KB\frac{\omega_n^2}{\omega_d}e^{-\sigma t}\sin(\omega_d t)$$

$$\dot{y}(t_p) = 0 \quad \Longrightarrow \quad t_p = \frac{\pi}{\omega_d} \quad \text{(first maximum)}$$

$$M_p = \frac{y(t_p) - KB}{KB} = e^{-\sigma\pi/\omega_d} = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)$$




Example: ζ = 0.5; first maximum and peak gap

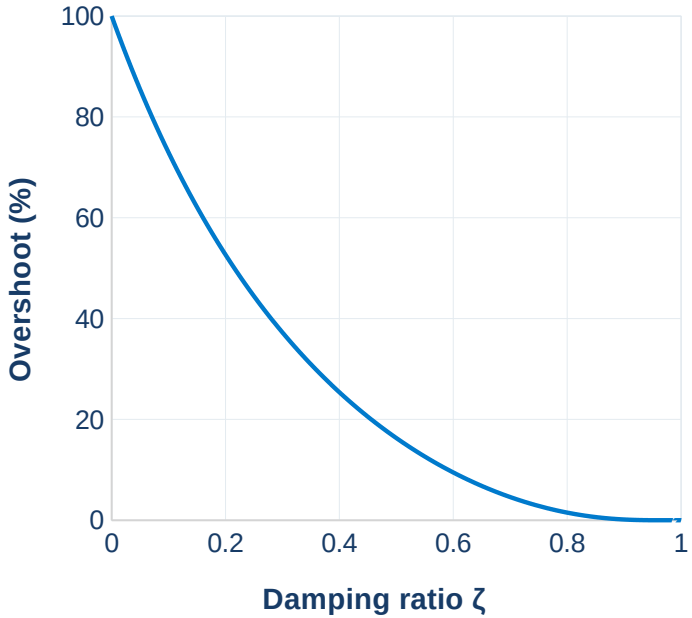
Percentage overshoot is $100M_p$. These expressions require a standard second-order transfer function with a **constant numerator, zero initial state** and $0 < \zeta < 1$.

# Overshoot at fixed damping ratio



$$M_p = 100 \cdot exp\left[-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right]\%$$

At fixed $\zeta$, changing $\omega_n$ rescales time without changing percentage overshoot.




Standard underdamped second-order model; constant numerator; zero initial state.

# Rise and settling times



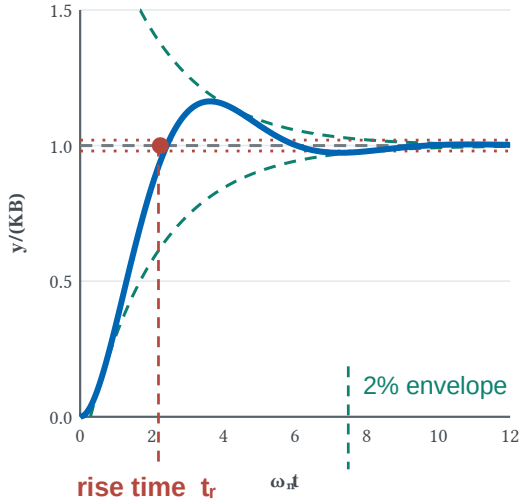
*An exact first crossing and a conservative error envelope*

$$y(t_r) = KB \Longrightarrow \cos(\omega_d t_r) + \frac{\sigma}{\omega_d}\sin(\omega_d t_r) = 0$$

$$\phi = \arccos\zeta \Longrightarrow t_r^{(0--100\%)} = \frac{\pi - \phi}{\omega_d} \qquad (0 < \zeta < 1)$$

$$\frac{|KB - y(t)|}{KB} \leq \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}}$$

$$t_{s,\mathrm{env}}(\varepsilon) = \frac{\ln\left[1/\left(\varepsilon\sqrt{1-\zeta^2}\right)\right]}{\sigma}$$




Blue: response Green: envelope Red: ±2% band

For the 2% band, use $t_s \approx 4/(\zeta\omega_n)$; for 5%, use $\approx 3/(\zeta\omega_n)$. **These are design estimates**. The actual last crossing depends on oscillation phase; verify it from the simulated curve.

# Live demo: damping and response speed

OPEN LIVE DEMO ↗




Course home: Demo 2.

START $K = 1$ , $\zeta = 0.5$ , $\omega n = 2$ rad/s. Unit step.

1 Keep $\omega n = 2$ . Try $\zeta = 0.2$ , 0.5, 1 and 1.5. Compare overshoot and poles.

2 Restore $\zeta = 0.5$ . Set $\omega n = 4$ . Which times change?

## MATLAB alternative (Control System

```
K=1; zeta=0.5; wn=2; s=tf('s');
G=K*wn^2/(s^2+2*zeta*wn*s+wn^2);
figure; step(G,0:0.01:10); grid on
figure; pzmap(G); grid on
stepinfo(G,RiseTimeLimits=[0 1])
```

CHECK At $\zeta = 0.5$, $\omega_n = 2$: $M_p = 16.3\%$, $t_p = 1.81$ s. Doubling $\omega_n$ at fixed $\zeta$ halves the time scale but leaves percentage overshoot unchanged.

PART 2

# Identification and model limits

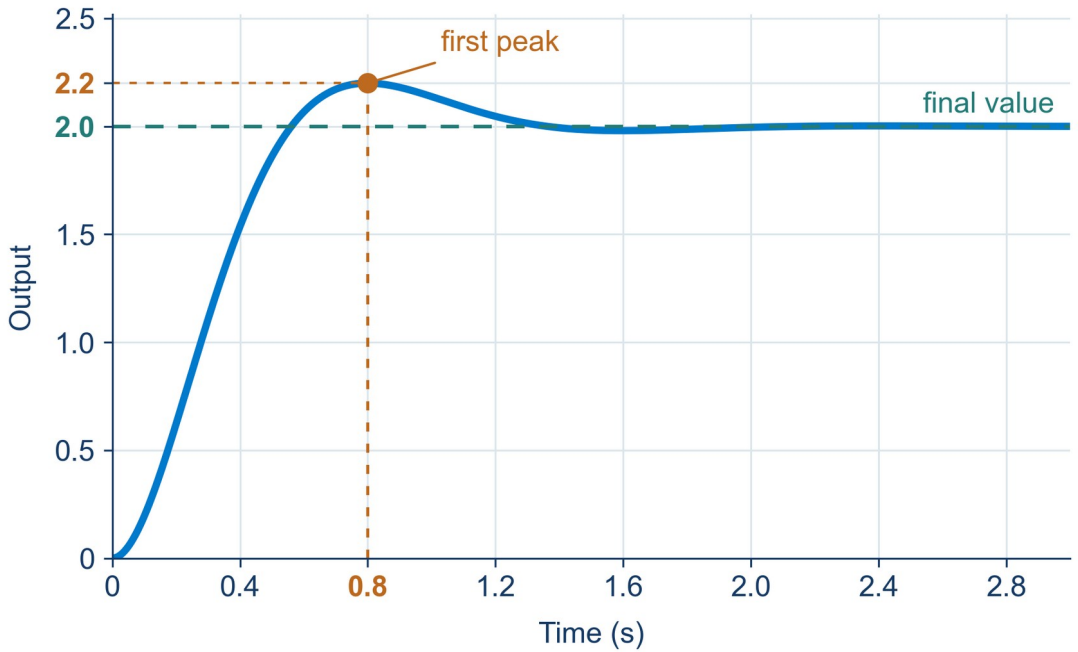
01 Read the response

02 Transient identification

03 Dominant poles and zeros

# Step-Response Identification

QUESTION




**Read directly from the curve**

1 Final value

2 First peak

3 Time of first peak

Then estimate the gain, damping ratio and natural frequency.

# Worked: identification from a transient

EXAMPLE

*From the curve: final 2.0, first peak 2.2, peak time 0.8 s*

**Use the overshoot ratio 0.10, not the percentage number 10.**

$$M_p = \frac{2.2 - 2.0}{2.0} = 0.10, \quad \zeta = 0.591$$

$$\ln M_p = -\frac{\pi\zeta}{\sqrt{1-\zeta^2}}$$

$$(\ln M_p)^2 \left(1-\zeta^2\right) = \pi^2\zeta^2$$

$$\zeta = \frac{-\ln M_p}{\sqrt{\pi^2 + (\ln M_p)^2}}$$

$\omega_d = \frac{\pi}{0.8} = 3.927$ rad/s

$\omega_n = \frac{3.927}{\sqrt{1-0.591^2}} = 4.87$ rad/s

Unit-step DC gain = 2.

**The model**

Poles $\approx -2.88 \pm j3.93$

$$G(s) = 2 \cdot \frac{23.7}{s^2 + 5.76s + 23.7}$$

Predicted $t_s \approx \frac{4}{2.88} = 1.39$ s; compare with the simulated curve.

# Higher-order response as a sum of modes



A real-coefficient LTI denominator factors into real linear terms and irreducible quadratic terms. Partial fractions turn them into first- and second-order modes.

$$Y(s) = \frac{\Delta u\, G(s)}{s} = \frac{y_\infty}{s} + \sum_i \frac{A_i}{s - p_i} + \sum_j \frac{B_j (s - \alpha_j) + C_j \beta_j}{(s - \alpha_j)^2 + \beta_j^2}$$

$$y(t) = y_\infty + \sum_i A_i e^{p_i t} + \sum_j e^{\alpha_j t} \left( B_j \cos \beta_j t + C_j \sin \beta_j t \right)$$

Repeated poles add time factors; **zeros change the weight of each mode**. The sum-of-modes result assumes a linear time-invariant system.

# Dominant poles in a higher-order model



***The slow poles nearest the imaginary axis shape the visible response***

$$G(s) = \frac{250}{(s+10)(s^2+2s+25)}, p_1 = -10, \quad p_{2,3} = -1 \pm j\sqrt{24}$$

$$G_{\text{red}}(s) \approx \frac{25}{s^2+2s+25}, \zeta = 0.20, \quad \omega_n = 5 \text{ rad/s}$$

The complex pair decays more slowly than the −10 pole.

Approximate s+10 by 10. No zero nearly cancels the pair.

A dominant pole must have a visible contribution; a nearby zero may suppress it.

# First order and second order, side by side



| | First order $\frac{K}{\tau s+1}$ | Second order |
|---|---|---|
| **Parameters** | $\tau, K$ | $\zeta, \omega_n, K$ |
| **Overshoot** | never | when $\zeta < 1$ |
| **Settling** $(2\%)$ | $t_s \approx 4\tau$ | $t_s \approx \frac{4}{\zeta\omega_n}$ |
| **Speed set by** | $\tau$ | $\zeta\omega_n$ |
| **Identify from** | the 63.2 % point | overshoot and peak time |

PART 3

# Prediction and requirements

01 Check predictions in MATLAB

02 Translate requirements

# Predict, then simulate

EXAMPLE

*MATLAB Control System Toolbox checks the estimates*

**PREDICTION**

**For** $G = \frac{25}{s^2+4s+25}$

Overshoot about 25 %
Peak at 0.69 s
Settled by 2.0 s
Final value 1

**SIMULATION**

**MATLAB code**

```
G = tf(25, [1 4 25]);
step(G, 0:0.001:3); grid on
stepinfo(G,RiseTimeLimits=[0 1])
damp(G)      % ζ and ωn for each pole pair
```

# Response Requirements and Pole Constraints



| Requirement as stated | Becomes | Which constrains |
|---|---|---|
| **"No visible bounce"** | $M_p < 5\%$ | $\zeta > 0.69$ |
| **"Ready within half a second"** | $t_s < 0.5$ s | $\zeta\omega_n > 8$ |
| **"Must not oscillate at all"** | $\zeta \geq 1$ | no complex pair |
| **"Follow a 2 Hz command"** | bandwidth $> 12.6$ rad/s | $\omega_n$, via Lecture 5 |

**Requirements define an $s$-plane region.**


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